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File operation

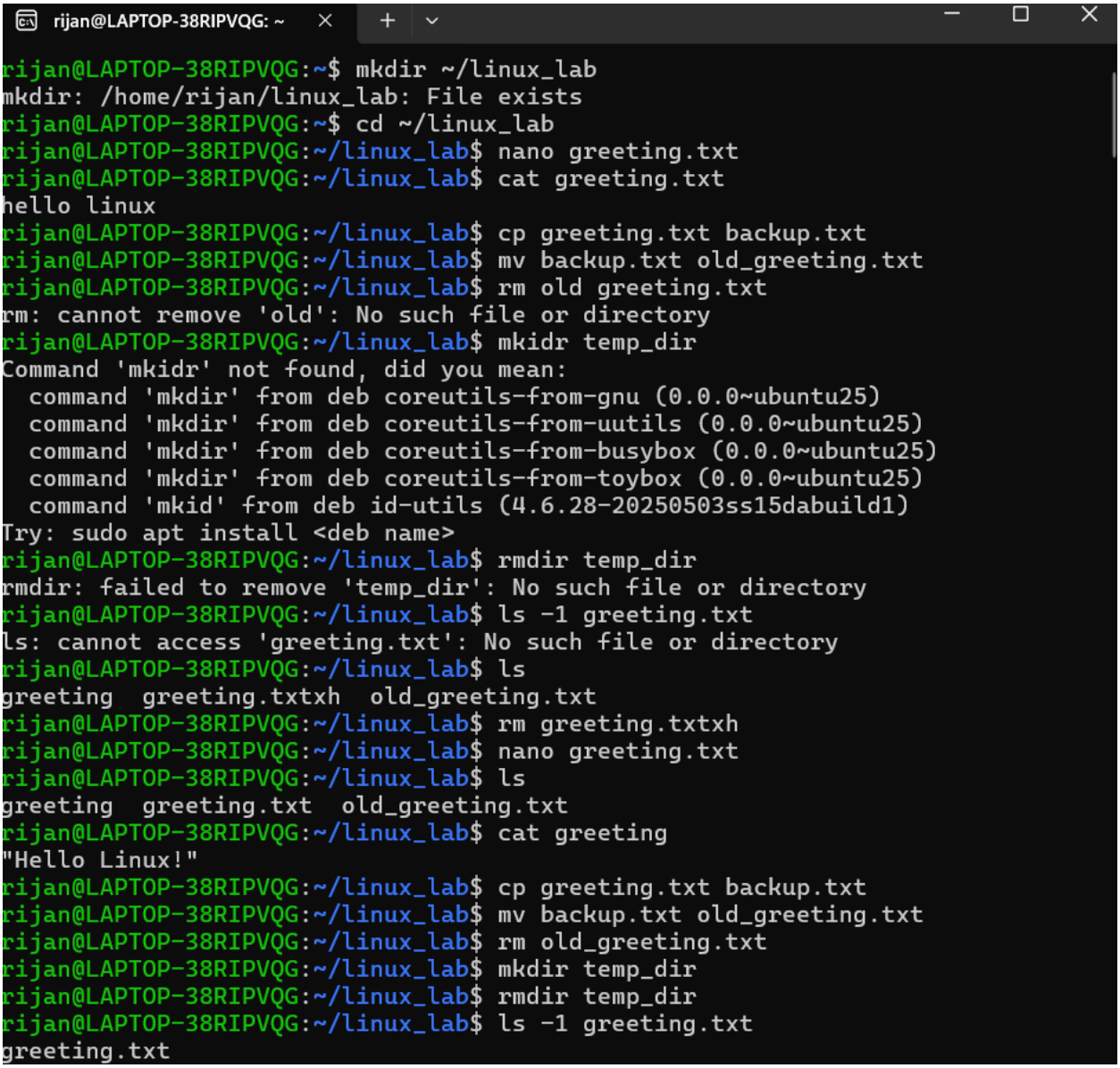


Figure 1: File commands

Summary
I set up a fresh Linux working directory and got hands-on with basic command-line file management. I wrote some text in greeting.txt using Nano, checked my work with cat, and practiced duplicating, renaming, and removing files with cp, mv, and rm. To round it out, I made and deleted a directory and used ls -l to check permissions.

APT Management

```
rijan@LAPTOP-38RIPVQG:~/linux_lab$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
67 packages can be upgraded. Run 'apt list --upgradable' to see them.
rijan@LAPTOP-38RIPVQG:~/linux_lab$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
```

Figure 2:Sudo apt upgrade

GCC verification

```
rijan@LAPTOP-38RIPVQG:~/linux_lab$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Figure 3: GCC Verification and Package management

Summary

Package Management & System Maintenance: Updated local APT repositories and managed package installations on Linux/Ubuntu systems.

Development Environment Setup: Installed build-essential toolchain, configuring core C/C++ build tools and the GCC compiler for local development.

Verification & Querying: Validated compiler installations using CLI flags (gcc --version) and queried APT repositories (apt search) to identify and manage Python 3 dependencies.

Compilation

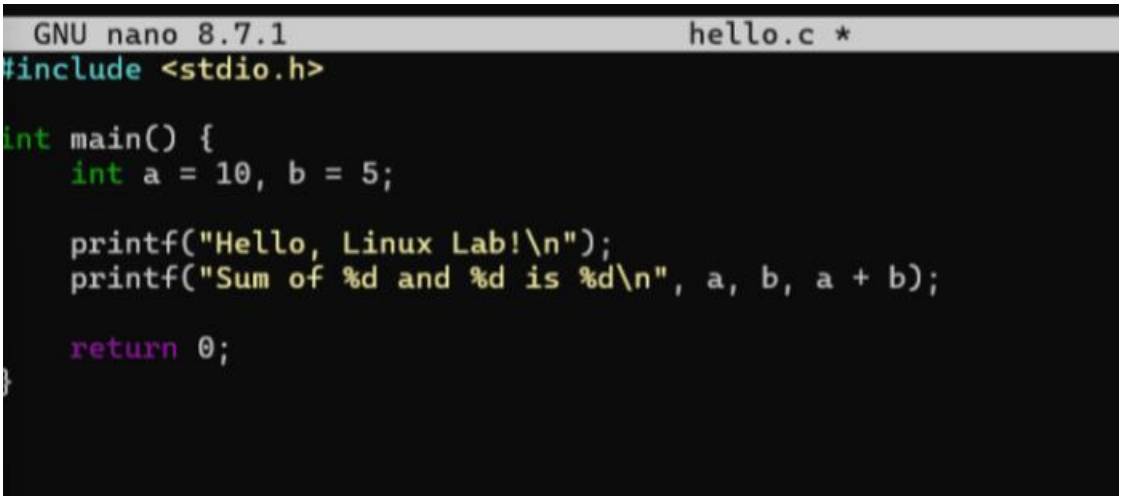


Figure 4:Source codes

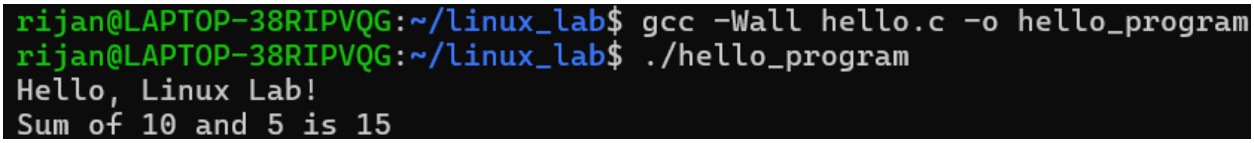


Figure 5: Compiling and running the C program

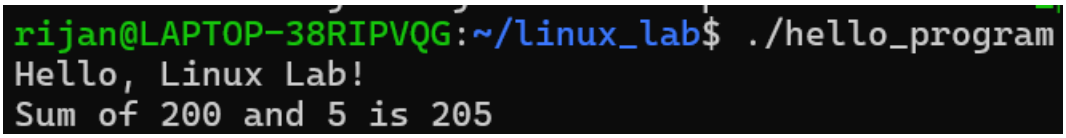
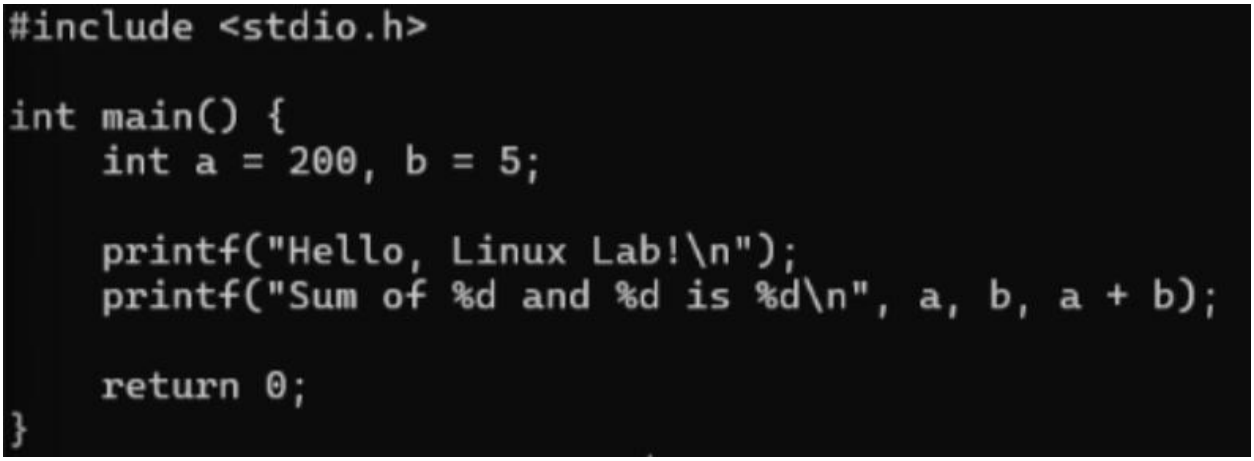


Figure 6:Modified and Recompiling the program



```c
#include <stdio.h>

int main() {
    int a = 200, b = 5;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
```

Figure 7: Final Verification of the lab

Summary

I wrote a C program and then compiled it and ran it. The program uses the input/output library called stdio.h. The program shows a message on the terminal screen. It also adds two numbers, which're 200 and 5. It uses the %d format specifier with the print f function to show the result of the addition.